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How this paper compares to your past papers

Based on 4 uploaded past papers

76%
76% match

Topic coverage 60% Format match 100%

Covered in this paper:

✓ Binomial distribution $B(n,p)$ — calculating exact probabilities and real-world applications

✓ Normal distribution — standardisation and probability calculations

✓ Normal approximation to binomial distribution with continuity correction and hypothesis testing

✓ Binomial distribution — application to real-world scenarios and Normal approximation with continuity correction

✓ Hypothesis testing — constructing null/alternative hypotheses and p-value interpretation, Confidence intervals

✓ Poisson distribution — exact probabilities and approximation considerations

Queued for next paper:

→ Discrete probability distributions — expected value $E(X)$ and variance $Var(X)$

AUSTRALIAN CAPITAL TERRITORY

Narrabundah College

what — Year

BSSS Mock Examination — Paper 1

Scope: SMO5 Statistics and Statistics Extension · mid unit test

TOTAL MARKS

56

TIME ALLOWED

55 minutes

PERMITTED
MATERIALS

Scientific
calculator,
ruler

Instructions to candidates

- Write in black or blue pen only
- Show all working clearly for full marks
- Scientific calculator permitted
- Round answers to 3 decimal places where appropriate
- Questions — 56 marks

Sign out

Full name	Teacher	Date

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Questions

Answer ALL questions. Show all working clearly. Marks allocated are shown in brackets.

56 marks

- Q1** A pharmaceutical company is testing a new drug for treating migraines. In clinical trials, it is known that 15% of patients experience side effects. A random sample of 20 patients is selected. [9 Marks – 1, 2, 3, 3]
- a)** State the distribution of X , the number of patients experiencing side effects, and identify the values of n and p . [1 mark]
- b)** Calculate $P(X = 3)$ using appropriate calculator notation. Show your working. [2 marks]
- c)** Find the probability that at most 2 patients experience side effects. Use appropriate calculator syntax and give your answer to 3 decimal places. [3 marks]
- d)** Calculate the expected number of patients who will experience side effects and the variance of this distribution. [3 marks]

9 marks	Binomial distribution $B(n,p)$ — calculating exact probabilities and real-world applications	Show answer ↓
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Q2 The heights of male students at a college are normally distributed with mean $\mu = 178$ cm and standard deviation $\sigma = 8$ cm. [9 Marks – 2, 2, 2, 3]

a) A student is 190 cm tall. Calculate the standardised score (z-score) for this height.
[2 marks]

b) Using your z-score from part (a), find the probability that a randomly selected male student is taller than 190 cm. Give your answer to 4 decimal places. [2 marks]

c) Find the probability that a randomly selected male student has a height between 170 cm and 186 cm. [2 marks]

d) Using calculator syntax, find the height below which 10% of male students fall. Show your working clearly. [3 marks]

9 marks Normal distribution — standardisation and probability calculations [Show answer](#) ↓

Q3 A manufacturing plant produces light bulbs where the probability of a bulb being defective is 0.08. Quality control tests a batch of 150 bulbs. [10 Marks – 2, 2, 3, 3]

a) State why a normal approximation to the binomial distribution is appropriate here. Check the necessary conditions. [2 marks]

b) Calculate the mean and standard deviation of the normal approximation. Use

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these to state the approximate distribution.

[2 marks]

c) Using the normal approximation with continuity correction, find the probability that exactly 10 bulbs are defective. Show all steps. [3 marks]

d) The quality control manager claims the defect rate is exactly 8%. A sample of 150 bulbs is tested and 16 defective bulbs are found. Construct a hypothesis test at the 5% significance level to determine if there is evidence the true defect rate differs from 8%. State your hypotheses, calculate the p-value, and conclude. [3 marks]

10 marks Normal approximation to binomial distribution with continuity correction and hypothesis testing [Show answer](#) ↓

Q4 A pharmaceutical company is testing a new antihistamine drug. In clinical trials, it is known that 12% of patients experience mild side effects. The company conducts a trial with 150 randomly selected patients.

a) Calculate the probability that exactly 18 patients experience mild side effects using the binomial distribution $B(150, 0.12)$. Give your answer to 4 decimal places. [2 marks]

b) Find $P(X \leq 20)$ where X represents the number of patients experiencing side effects. [2 marks]

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c) The company wants to use a Normal approximation to the binomial distribution. Calculate the mean μ and standard deviation σ for this approximation. [2 marks]

d) Using the Normal approximation with continuity correction, find $P(17 \leq X \leq 21)$. Show all working including the standardisation formula $Z = (X - \mu)/\sigma$. [3 marks]

9 marks Binomial distribution — application to real-world scenarios and Normal approximation with continuity correction [Show answer](#) ↓

Q5 A survey of 200 students at a college found that 68 are left-handed. Previous research suggests the population proportion of left-handedness is 30%.

a) Construct appropriate null and alternative hypotheses to test whether the sample provides evidence that the proportion of left-handed students differs from 30%. [2 marks]

b) Calculate the sample proportion $\hat{p}$ of left-handed students. [1 mark]

c) Perform a hypothesis test at the 5% significance level using the binomial distribution. Calculate $P(X \geq 68)$ where $X \sim B(200, 0.30)$, then explain whether there is sufficient evidence to reject H_0 . [3 marks]

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d) Calculate a 95% confidence interval for the population proportion of left-handed students. Use the formula: $CI = \hat{p} \pm 1.96\sqrt{\hat{p}(1-\hat{p})/n}$. Give your answer to 3 decimal places. [3 marks]

9 marks Hypothesis testing — constructing null/alternative hypotheses and p-value interpretation, Confidence intervals

Show
answer
↓

Q6 The number of customer complaints received per week at a technical support centre follows a Poisson distribution with mean $\lambda = 5.4$.

a) Calculate $P(X = 6)$ where $X \sim \text{Po}(5.4)$, giving your answer to 4 decimal places. [2 marks]

b) Find $P(X < 4)$, giving your answer to 4 decimal places. [2 marks]

c) Calculate $E(X)$ and $\text{Var}(X)$ for this Poisson distribution. [2 marks]

d) The support centre receives complaints over 4 weeks. Find the probability that the total number of complaints over this 4-week period is exactly 22. [2 marks]

e) Explain why a Normal approximation would be appropriate for the 4-week total, and use it to find $P(X < 20)$ where X is the total complaints over 4 weeks. Use continuity correction. [2 marks]

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10 marks	Poisson distribution — exact probabilities and approximation considerations	Show answer ↓
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